A distorted tower sieve for odd covering systems, checked in Lean

Ryan Wu Claude Fable 5.1

November 2026 (draft v0.2)

Abstract

A covering system is a finite family of residue classes whose union is $\mathbb{Z}$. Erdős and Selfridge asked whether one can have all moduli odd, distinct, and greater than one; the question is open. We prove, with every step checked by the Lean 4 kernel and no compiled evaluation, that *the least common multiple of the moduli of such a system is at least* $1\,527\,701\,175$. Three ingredients are formalised. First, a sieve certificate: removing one residue class for each prime-power modulus $p^j \mid L$ leaves a Chinese-remainder product set, and a union bound over the composite classes inside it gives a rational $T(L)$, sharp at $L = 12, 80$, with $T(L) \geq 1$ for every covering number; its exponent-free limit is the 1986 condition of Berger, Felzenbaum and Fraenkel, and it yields that every odd covering needs at least five primes. Second, a *distorted* version: at each prime we condition on the complement of the prime-power classes and then apply the distortion of Balister, Bollobás, Morris, Sahasrabudhe and Tiba to the mixed classes; this breaks the barrier at $11\,486\,475$ shared by all first-order bounds we know. Third, prime replacement for arbitrary exponents, which reduces the problem to the 1638 moduli of the form $3^a5^b7^c\cdots$ below the bound, each closed by an explicit certificate. We place the results against Mian–Siddique, Hackmon, McNew–Setty and Schroeder, stating for each claim whether it is proved, machine-checked, numerical, or unchecked.

1 The question, and a twelve-line example

Can the integers be covered by finitely many arithmetic progressions with pairwise distinct odd differences? Nobody knows. With even differences allowed the answer is yes, and the smallest example is as old as the subject:

$$0 \pmod 2, \quad 0 \pmod 3, \quad 1 \pmod 4, \quad 1 \pmod 6, \quad 11 \pmod{12},$$

which covers $\mathbb{Z}$ with moduli dividing 12. Odd moduli resist. The sum of reciprocals of the moduli must be at least 1 (each class has density $1/m$), so $\sum_{d \mid L, d>1} 1/d \geq 1$ for the lcm L: L must be *abundant* or perfect, and the least odd such number is 945. This is the only obstruction that uses no structure at all, and it is far from the truth.

A **covering system** is a finite family $\{a_i \pmod{m_i}\}$ with $\bigcup_i (a_i + m_i\mathbb{Z}) = \mathbb{Z}$; it is **distinct** if the m_i are pairwise different, and **odd** if every m_i is odd. Moduli are always > 1. We write $L = \operatorname{lcm}(m_1, \ldots, m_k)$ and say L is a **covering number** if some distinct covering system has lcm L (equivalently, has all moduli among the divisors of L). Erdős and Selfridge's question is whether an odd covering number exists.

Question 1 (Erdős–Selfridge; Erdős Problem 7). Is there a distinct covering system all of whose moduli are odd?

We do not answer it. We prove, with every step of the mathematics checked by the Lean proof kernel, that an odd covering number would have to be large, and we say precisely how large and

under which trust assumptions. The proof is a sieve so simple that a proof kernel can evaluate it a few hundred thousand times.

Here is the idea on $L = 12$. The moduli $2, 4$ each occur at most once, so their two classes remove at most $2 + 1 = 3$ of the 4 residues modulo 4, leaving at least $u_2 = 1$; the modulus 3 removes one of three residues, leaving $u_3 = 2$. By the Chinese remainder theorem the residues modulo 12 that survive both form a product set U of size at least $1 \cdot 2 = 2$. Every point of U must be covered by a class with composite modulus, here 6 or 12. A class modulo 6 meets U in at most $\min(2, 1) \cdot \min(1, 2) = 1$ point; a class modulo 12 in at most 1. So two composite classes can cover at most 2 points, which is exactly $|U|$: the bound is $T(12) = 1/1 \cdot 1/2 + 1/1 \cdot 1/2 = 1$, and the example above shows it is attained. Had $T(12)$ come out below 1, there would be no covering with lcm 12. That is the whole method.

Section 2: the certificate. The sieve above, written for general L, is the tower certificate $T(L)$; its cruder cousin, the φ-certificate, sieves only the prime moduli. Both are proved.

Section 3: what it sharpens. Letting every exponent go to infinity turns $T(L) < 1$ into the condition of Berger, Felzenbaum and Fraenkel. So the certificate is the finite form of a thirty-year-old criterion, and Hackmon's recent four-prime theorem is the same criterion at $\{3, 5, 7, 11\}$.

Section 4: what is checked. The Lean development, the four certified theorems, and what each trusts.

Section 6: where this sits. Prior and concurrent work, including a Zenodo preprint claiming a bound about six times larger than ours, which we have not been able to rebuild, and the inclusion–exclusion lemma of McNew–Setty, which has no written proof.

2 The certificate

Fix $L = \prod_p p^{e_p}$. For each prime $p \mid L$ put

$$u_p \;=\; p^{e_p} - \bigl(1 + p + \cdots + p^{e_p - 1}\bigr),$$

the number of residues modulo p^{e_p} that are guaranteed to survive when one class modulo $p^{e_p - j}$ is removed for each $j = 0, \ldots, e_p - 1$ (a class modulo $p^{e_p - j}$ is p^j residues modulo p^{e_p}). Note $u_p \geq p^{e_p}(p-2)/(p-1)$. Write $v_p(d)$ for the exponent of p in d and $\omega(d)$ for the number of distinct primes dividing d.

Definition 2 (Tower sum)**.**

$$T(L) \;=\; \sum_{\substack{d \mid L \\ \omega(d) \geq 2}} \prod_{p \mid L} \frac{\min\bigl(p^{\,e_p - v_p(d)},\; u_p\bigr)}{u_p}.$$

In words: for each composite divisor d, and each prime p, take the fraction of the surviving residues modulo p^{e_p} that one class modulo $p^{v_p(d)}$ can hit, and multiply over primes. For $p \nmid d$ the factor is $\min(p^{e_p}, u_p)/u_p = 1$, so only primes dividing d contribute.

Theorem 3 ((Tower certificate))**.** *If L is the lcm of a covering system with distinct moduli > 1, then $T(L) \geq 1$.*

Proof. Reduce all classes modulo L; the family covers $\mathbb{Z}/L$.

Step 1: sieve the towers. For each $p \mid L$ the moduli $p, p^2, \dots, p^{e_p}$ each occur at most once. Their classes remove at most $p^{e_p-1} + \cdots + 1$ residues modulo p^{e_p}, so some set $A_p \subseteq \mathbb{Z}/p^{e_p}$ with $|A_p| \geq u_p$ is disjoint from all of them. Let U be the set of $n \in \mathbb{Z}/L$ with $n \bmod p^{e_p} \in A_p$ for every p. By the Chinese remainder theorem $|U| = \prod_p |A_p|$, and every point of U is covered by some class whose modulus d has $\omega(d) \geq 2$.

Step 2: count one composite class inside U. Fix $c \pmod d$ with $d \mid L$. A point $n \in U$ lies in it exactly when, for every p, the residue $n \bmod p^{e_p}$ lies in A_p and is $\equiv c \pmod{p^{v_p(d)}}$. These conditions are independent across primes, so

$$\big|\{c \bmod d\} \cap U\big| \;=\; \prod_p \big|\{x \in A_p : x \equiv c \pmod{p^{v_p(d)}}\}\big| \;\leq\; \prod_p \min\!\big(p^{\,e_p - v_p(d)},\ |A_p|\big),$$

because a class modulo $p^{v_p(d)}$ has $p^{e_p - v_p(d)}$ elements modulo p^{e_p}.

Step 3: union bound, then normalise. The composite classes cover U, so $|U| \leq \sum_{\omega(d)\geq 2} |\{c_d\} \cap U|$. Divide by $|U| = \prod_p |A_p|$ and use that $s \mapsto \min(a, s)/s$ is non-increasing together with $|A_p| \geq u_p$:

$$1 \;\leq\; \sum_{\omega(d)\geq 2} \prod_p \frac{\min(p^{e_p - v_p(d)}, |A_p|)}{|A_p|} \;\leq\; \sum_{\omega(d)\geq 2} \prod_p \frac{\min(p^{e_p - v_p(d)}, u_p)}{u_p} \;=\; T(L). \qquad \square$$

The same argument with only the prime moduli sieved (take A_p to be the complement of one class modulo p, so $|A_p| = \varphi(p^{e_p})$) gives a weaker but even simpler statement, since then the ratio in Step 3 collapses: for $p \mid d$ the p-factor is at most $p^{e_p - v_p(d)} / \big(p^{e_p - 1}(p-1)\big) = 1/\big(p^{v_p(d)-1}(p-1)\big)$, and the product over $p \mid d$ is $1/\varphi(d)$.

Theorem 4 ($(\varphi$-certificate$)$)**.** *If L is a covering number then $\sum_{d \mid L,\ d \text{ composite}} 1/\varphi(d) \geq 1$.*

For squarefree L the two certificates coincide; in general Theorem 3 is at least as strong.

Example 5 (Sharpness)**.** $T(12) = 1$ (computed above) and $T(80) = 1$; both 12 and 80 are covering numbers. So no constant below 1 can replace the 1 in Theorem 3.

Example 6 (Where each certificate first fails on odd numbers)**.** Scanning odd non-deficient L in increasing order, the φ-certificate first fails to exclude $675\,675 = 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$, and the tower certificate first fails at

$$6\,891\,885 = 3^4 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17, \qquad T = 1.0098\ldots$$

These are numerical facts, cross-checked by two independent implementations (Python with exact rationals, and the Lean evaluation of Section 4). Whether 6 891 885 is a covering number is not known to us; the certificate simply says nothing about it.

Remark 7. Theorem 3 is a first-order union bound inside U. It ignores that two composite classes overlap. We tested numerically the natural refinement, a pairwise-coprime inclusion–exclusion inside U in the manner of McNew–Setty; it excludes 6 891 885 and first fails at $11\,486\,475 = 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, which is also where McNew–Setty's own inequality first fails on odd numbers. We have not proved the refinement; see Section 6.

3 What the certificate sharpens

Berger, Felzenbaum and Fraenkel proved in the 1980s that the primes dividing a covering number N satisfy

$$\prod_{p\mid N}\frac{p-1}{p-2}\;-\;\sum_{p\mid N}\frac{1}{p-2}\;>\;2. \tag{1}$$

This condition sees only the prime support, not the exponents. Our certificate sees both, and (1) is what remains when the exponents are forgotten.

Proposition 8 ((Tower implies BFF))**.** *Let P be the set of primes dividing L. Then*

$$T(L)\;\le\;\prod_{p\in P}\frac{p-1}{p-2}\;-\;1\;-\;\sum_{p\in P}\frac{1}{p-2}.$$

Consequently, if L is a covering number then (1) *holds with $\ge$.*

Proof. For $p\mid d$ with $v=v_p(d)\ge 1$, using $u_p\ge p^{e_p}(p-2)/(p-1)$,

$$\frac{\min(p^{e_p-v},u_p)}{u_p}\;\le\;\frac{p^{e_p-v}}{u_p}\;\le\;\frac{p-1}{p-2}\,p^{-v},$$

and for $p\nmid d$ the factor is 1. Group the composite divisors by their support $S\subseteq P$, $|S|\ge 2$, and sum the geometric series in each exponent:

$$T(L)\;\le\;\sum_{|S|\ge 2}\;\prod_{p\in S}\frac{p-1}{p-2}\sum_{v=1}^{e_p}p^{-v}\;\le\;\sum_{|S|\ge 2}\;\prod_{p\in S}\frac{p-1}{p-2}\cdot\frac{1}{p-1}\;=\;\sum_{|S|\ge 2}\;\prod_{p\in S}\frac{1}{p-2}.$$

Finally $\sum_{S\subseteq P}\prod_{p\in S}\frac{1}{p-2}=\prod_{p\in P}\bigl(1+\frac{1}{p-2}\bigr)$, and removing the terms with $|S|\le 1$ gives the claim. □

Two readings. First, the certificate is strictly stronger than (1): for $P=\{3,5,7,11,13\}$ the left side of (1) is 2.14, so BFF excludes nothing with that support, while $T(L)<1$ for every odd L below 6 891 885 (Section 4), and in particular for every $L=3^a5^b7^c11^d13^e$ in that range. Second, Hackmon's "weight function" theorem (no odd covering supported on four primes, with ratio $41/45$ for $\{3,5,7,11\}$) is (1) at that prime set: $\prod\frac{p-1}{p-2}-1-\sum\frac{1}{p-2}=\frac{32}{9}-1-\frac{74}{45}=\frac{41}{45}$. His argument and ours are the same sieve; his bound uses the limiting factor $(p-2)/(p-1)$ in place of u_p/p^{e_p}. The exponent-free bound (1) forces only $\omega(L)\ge 5$ for odd coverings.

4 What is checked, and what each theorem trusts

Everything in Sections 2 and 3 is formalised in Lean 4 against Mathlib (toolchain `v4.22.0`); Proposition 8 is `towerSum_le_bff` in `BFF.lean`, and its corollary `no_covering_3_5_7_11` states that no covering system with distinct moduli has lcm $3^a5^b7^c11^d$ for any exponents $a,b,c,d\ge 1$, both with the standard three axioms. Pushing the same inequality through an induction on the primes in increasing order (the k-th smallest odd prime is at least the k-th of $3,5,7,11$, and D is monotone in each weight) gives the exponent-free statement `five_primes_of_odd_covering`: *every odd distinct covering system has at least five distinct primes dividing the lcm of its moduli*, again with the standard three axioms (`FourPrimes.lean`). Five is the limit of this method: as the exponents grow, $T(L)$ tends to the BFF value, which exceeds 1 already for $\{3,5,7,11,13\}$. The development is in the repository's `lean/Covering/` directory.

Statement. A covering of $\mathbb{Z}$ is a list of classes `Cls r d`; `IsCovering L S` packages: all moduli divide L, are > 1, are pairwise distinct, and the classes cover `range L`. The bridge lemma `isCovering_lcm` passes from a covering of $\mathbb{Z}$ with distinct moduli > 1 to `IsCovering` at $L = \mathrm{lcm}$ of the moduli, and `lcm_odd` records that the lcm of odd numbers is odd.

The two inequalities. `one_le_towerSum` and `one_le_phiSum` are Theorems 3 and 4. The CRT product count of Step 2 is `card_cls_inter_prodSet`, proved for arbitrary allowed residue sets by induction over the prime set. Their axioms are `propext`, `Classical.choice`, `Quot.sound`: the standard three.

Decidable certificate. `towerSum L` is defined with Mathlib's `Nat.divisors` and `primeFactors`, which a kernel cannot evaluate quickly. We therefore define a second, list-based evaluator (verified $\sqrt{L}$ divisor enumeration, trial-division primality, p-adic valuation by repeated division) and prove it equal to the Mathlib expression, so that `towerCheck L = true` implies $T(L) < 1$ by `no_covering_of_towerSum`. A chunked scan `scanChunk2 lo cnt` checks every odd L in a range, with an append lemma to glue chunks.

Results. Let $\mathcal{C}$ be any odd distinct covering of $\mathbb{Z}$ with moduli > 1 and L its lcm.

Theorem	Statement	Axioms
`five_primes_of_odd_covering`	$\omega(L) \geq 5$	standard three
`odd_covering_lcm_gt_6891884_kernel`	$L > 6\,891\,884$ (tower certificate)	standard three
`odd_covering_lcm_gt_1527701174`	$L > 1\,527\,701\,174$ (distorted tower)	standard three

"Standard three" is `propext`, `Classical.choice`, `Quot.sound`: every arithmetic step, including the evaluation of the certificates, is checked by the kernel (`decide +kernel`); nothing uses `native_decide`, and there is no `sorry`. The whole library builds from scratch in about seven minutes on two cores. The previous certified bound was $L > 10^4$ (Mian–Siddique, July 2026), whose prime-only CRT certificate stalls at 10 395; its authors name "charging prime-power towers" as the needed refinement, which is what $T(L)$ does.

5 Breaking the first-order barrier

Every first-order bound we know stops at the same number. The tower certificate first fails on odd numbers at 6 891 885; adding McNew–Setty's pairwise-coprime inclusion–exclusion inside U, McNew–Setty's own inequality, and the distorted sieve of Balister, Bollobás, Morris, Sahasrabudhe and Tiba (BBMST) evaluated as written all fail at

$$11\,486\,475 = 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17.$$

The two ideas that work, conditioning on the tower and distorting, act on different classes, and combining them is stronger than either.

The certificate. List the primes of L in some order $p_1, \ldots, p_n$, write $Q_i = p_1^{e_1} \cdots p_i^{e_i}$, and choose caps $\delta_i \in [0, \frac{1}{2}]$. Put $g_k(v) = \min(p_k^{e_k - v}, u_k)/u_k$, $G_k = \sum_{v=1}^{e_k} g_k(v)$, and let ν be the multiplicative

function with $\nu(p_k^v) = g_k(v)/(1-\delta_k)$ for $v \geq 1$. Define

$$M_i^{(1)} = G_i \sum_{\substack{m \mid Q_{i-1} \\ m>1}} \nu(m), \qquad M_i^{(2)} = G_i^2 \sum_{\substack{m,m' \mid Q_{i-1} \\ m,m'>1}} \nu(\mathrm{lcm}(m,m')), \qquad \eta = \sum_i \min\Big(M_i^{(1)}, \frac{M_i^{(2)}}{4\delta_i(1-\delta_i)}\Big),$$

dropping the second option when $\delta_i = 0$. With every $\delta_i = 0$ one recovers $\eta = T(L)$.

Theorem 9 ((Distorted tower certificate))**.** *If $\eta < 1$ for some order and some caps, no covering system with distinct moduli > 1 has lcm L.*

Proof sketch. Each modulus d belongs to the stage i of the last prime $p_i \mid d$; write $d = mp_i^j$ with $m \mid Q_{i-1}$, and call it pure if $m = 1$ and mixed otherwise. We build weights P_i on $\mathbb{Z}/Q_i$ one stage at a time, writing $\mathbb{Z}/Q_i = \mathbb{Z}/Q_{i-1} \times \mathbb{Z}/p_i^{e_i}$ and calling $\{x\} \times \mathbb{Z}/p_i^{e_i}$ a fibre. In each fibre we first put the uniform weight on the residues A_i missed by the pure classes ($|A_i| \geq u_i$, as in Theorem 3); then, if $\alpha_i(x)$ is the fraction of A_i hit by mixed classes, we apply the BBMST rule: hit points are multiplied by $\max\{0, (\alpha-\delta)/(\alpha(1-\delta))\}$, the others by $\min\{1/(1-\alpha), 1/(1-\delta)\}$. This preserves fibre masses, gives the pure classes weight 0, gives the mixed classes weight $\max(0,\alpha-\delta)/(1-\delta) \leq \min(\alpha, \alpha^2/(4\delta(1-\delta)))$ per fibre, and gives every class $b \bmod m$ weight at most $\nu(m)$. Bounding $\alpha_i(x)$ by a sum over mixed classes and using the class bound for the previous stage gives $E\,\alpha_i \leq M_i^{(1)}$ and $E\,\alpha_i^2 \leq M_i^{(2)}$; distinctness of moduli enters only here. The covered set therefore has total weight at most η, while a covering would give it weight 1. □

The sums over divisors factor, so η costs $O(\omega(L)^2)$ operations. The formal proof is `Hybrid.one_le_eta` in `Hybrid/Main.lean`. It is stated for any family with distinct moduli > 1 dividing Q_n; the three invariants (mass, class bound, covered mass) are proved together by induction over the stages.

Sanity check. A sound certificate must never exclude a real covering number. Minimised over every prime order and every δ on the grid $k/40$, $\eta \geq 1$ on all 94 primitive covering numbers below 10^6 in McNew–Setty's table, with equality at 12, 80 and 53 248.

Prime replacement. If a covering with distinct moduli dividing Mq^e exists and $p \leq q$ are primes coprime to M, then one exists with moduli dividing Mp^e (`replace_one`; Simpson–Zeilberger, for arbitrary exponents). The map $\mathbb{Z}/p^e \to \mathbb{Z}/q^e$ that keeps base-p digits as base-q digits pulls each class modulo q^v back to the empty set or a class modulo p^v; with the identity on the M-coordinate this pulls the whole covering back, and distinct moduli stay distinct. Iterating (`reduce_to_shaped`), an odd covering with lcm L gives one with a no-larger modulus of the form $3^{a_1}5^{a_2}\cdots p_k^{a_k}$, all $a_i \geq 1$. There are 1638 such moduli below 1 527 701 175. A generator over the explicit list $3, 5, \ldots, 47$ enumerates them, with a proof that it misses none, and each one carries a certificate (prime order and caps $\delta_i = k/200$) that the kernel checks. The worst has $\eta = 0.99771$.

The next barrier. Scanning all odd $L \leq 10^{10}$ with $\sigma(L) \geq 2L$ (about 2×10^7 numbers), the distorted certificate fails on 20 of them, the smallest being

$$1\,527\,701\,175 = 3^3 \cdot 5^2 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19, \qquad \eta = 1.014.$$

Using moments up to order four instead of two gives no improvement.

6 Where this sits

We checked the literature on 2 November 2026 (SciSpace and alphaXiv; Elicit's API was not available to us) and list every item that bears on the claims above, with its status.

Density bound (folklore; Rafik, Zenodo, Jan. 2026). $\sum_{d\mid L,d>1} 1/d \geq 1$, hence odd $L \geq 945$. Strictly weaker than Theorem 4.

Berger–Felzenbaum–Fraenkel (1986–88). Condition (1). Recovered as the exponent-free limit of Theorem 3 (Proposition 8).

Simpson–Zeilberger (1991), Guo–Sun (2005), BBMST (2022). Squarefree odd case: lcm needs ≥ 18, then ≥ 22 primes; then impossible. Our scans concern arbitrary exponents, where the problem is open.

Hough–Nielsen (2019). Every distinct covering has a modulus divisible by 2 or 3, so $3 \mid L$ for an odd covering. Used implicitly: our scans range over all odd L, so nothing is assumed.

McNew–Setty (arXiv 2507.23041). Classify covering numbers to 10^6 with a MIP solver. Their Lemma 4.10 bounds the covered proportion by $\sum_{S \text{ pairwise coprime}} (-1)^{|S|+1}/\operatorname{lcm} S$, with the proof indicated as "follows likewise by inclusion–exclusion". *Status: unproved in writing.* The inequality is not pointwise: for moduli $\{2, 4, 3, 9\}$ with nested classes, a point in all four classes contributes $1-4+4=1$ to the alternating count of covered points where at most 1 is allowed, but 0 to the uncovered side, so any proof must average. It is a Shearer-type bound: pairwise-coprime subsets are the independent sets of the "share a prime" graph, and Shearer's theorem proves the lemma whenever $(1/m)_m$ lies in the Shearer region of that graph. We found no counterexample in 446 exactly solved instances, with many cases of equality, but every solvable instance was inside the Shearer region; whether $D(6\,891\,885)$ is inside it we do not know. On odd numbers the lemma, if true, first fails at 11 486 475.

Mian–Siddique (arXiv 2607.25628, July 2026). $L > 10^4$ in Lean, prime-only CRT certificate. Superseded by the first row of the table.

Hackmon (Zenodo, Apr. 2026). The tower sieve with the limiting factor, per prime support; four-prime theorem. Same method, seven months earlier, not formalised; equals (1) at $\{3, 5, 7, 11\}$ by Proposition 8.

Schroeder (Zenodo, Sept. 2026). Claims $\omega(L) \geq 9$ for every odd distinct covering, hence $L \geq 9\,704\,539\,845$, with a Lean 4 package (toolchain `v4.32.0-rc1`; no `sorry` or `native_decide` found by inspection; standard definition of an odd distinct covering). *Status: unchecked.* We could not rebuild the package in our environment. If it holds, it exceeds our bound 1 527 701 175 by a factor of about 6; our certificates are simpler and our proof uses no compiled evaluation, and its own Corollary 1.3 uses the squarefree case of Theorem 4.

What is next. In order: rebuild Schroeder's package and print its axioms (his claimed 9.7×10^9 is 6× ours, and his "conditional reweighting" should be compared with Theorem 9); push past 1 527 701 175 by treating the first mixed stage $\{3, 5\}$ exactly instead of by a union bound; and look for a structural statement (a bound on $\omega(L)$ beyond five) from the distorted certificate, as Theorem 9 is the first tool here that handles the mixed classes without inclusion–exclusion.

Reproducibility. Repository: `https://github.com/rw762-sketch/Open-Math-problem`. `lake build` rebuilds the Lean development from the Mathlib cache; `lake env lean Covering/Axioms.lean` prints the axioms of each headline theorem; `tower_certificate.py N` and `phi_certificate.py N` are the independent Python scans.